

Maitreya Mohapatra

AI Engineer



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📅 **Available:** full-time from 15 Aug 2026

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🇮🇳 **Nationality:** Indian

🐙 **GitHub:** github.com/mimo1999

Professional Summary

AI Engineer with 3+ years building production AI/ML systems across fintech, healthcare, and industrial R&D. Hands-on with agentic architectures (LangGraph, MCP), retrieval-augmented generation, and deploying ML services behind CI/CD. Co-inventor on two US patent applications for ML-based financial forecasting. Completing M.Sc. in Artificial Intelligence at FAU Erlangen-Nürnberg.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

10/2023 – 08/2026

Master of Science in Artificial Intelligence (*graduating August 2026*)

Grade: 1.6 (German scale, 1.0 = best) · Thesis: Multimodal AI for Clinical Voice Diagnostics

Kalinga Institute of Industrial Technology, Bhubaneswar (KIIT)

06/2017 – 06/2021

Bachelor of Technology in Computer Science and Engineering

Grade: 1.5 (German scale equivalent, 1.0 = best)

Work Experience

Working Student, R&D DI

04/2025 – present

Siemens Healthineers | Erlangen, Germany

Tech stack: Python, Azure DevOps, Slurm, Podman, Linux, Pytest

- Designed an MCP-based integration concept for orchestrating Questa OneSpin RTL validation through FuseSoC workflows.
- Redesigned Azure DevOps CI/CD pipelines into a parallel execution framework that dispatched independent hardware build flows from a single repository checkout, reducing CI runtime by ~20% and manual intervention by ~60%.
- Performed resource and compute utilization analysis of containerized (Podman) FPGA simulation workloads, validating Slurm-based offloading as a scalable alternative to the existing CI execution model.
- Architected a hybrid Azure DevOps–Slurm workflow that offloaded FuseSoC jobs to dynamically allocated HPC nodes, improving compute utilization by ~50% and reducing job turnaround by ~35%.
- Improved the reliability of FuseSoC-based FPGA development by implementing Pytest regression tests, resolving test-case bugs, and validating new package management features.

Teaching Assistant, Deep Learning

11/2024 – 03/2025

FAU Erlangen-Nürnberg | Erlangen, Germany

Tech stack: PyTorch, Python

- Coached students through implementing and debugging deep learning models (CNNs, RNNs, regularization, optimization) in PyTorch.
- Resolved student queries by explaining core concepts and techniques covered in the Deep Learning course.

Associate Software Engineer, Data Science

06/2021 – 08/2023

HighRadius | Bhubaneswar, India

Tech stack: Python, SQL, Snowflake, Jenkins, Docker, AWS

- Delivered ML-powered treasury forecasting solutions for 8 enterprise accounts, achieving 90%+ first-week and 95%+ one-year cumulative forecast accuracy by designing payroll cash-flow forecasting models combining clustering and regression.
- Improved forecasting accuracy by 20–30% across production accounts by developing a customer behavior-driven feature engineering framework that generalized across heterogeneous client datasets.
- Reduced client implementation time by 10–15% by building scalable SQL and Snowflake ingestion pipelines that standardized data from multiple ERP systems into a unified modeling framework.
- Led delivery of AI forecasting solutions across 8 enterprise accounts by serving as the primary technical point of contact, mentoring 10+ interns, and translating business requirements into production ML systems.
- Reduced annual AWS infrastructure costs by ~20% by optimizing ML data pipelines using multiprocessing and compute-efficient pipeline design.

- Reduced feature-selection runtime 40% by replacing exhaustive grid search with an auxiliary feature-importance pre-filter.
- Cut manual effort in forecast loading 70% by automating a legacy UI workflow with credential-parameterized cloud scripts.

Projects

Research Swarm: Autonomous Multi-Agent Research System [GitHub](#)

LangGraph, LangChain, RAG, ChromaDB, Pydantic, Ollama

- Built a LangGraph multi-agent research system with parallel sub-question dispatch, per-session RAG (ChromaDB + BGE-small + cross-encoder reranker), and a critic|fact-checker|writer pipeline with HITL review; runs fully local via Ollama.
- Engineered for production: Pydantic structured outputs, SSRF-protected URL fetching, per-session LLM budget guard, and SqliteSaver checkpoint, validated on BEIR retrieval benchmarks (nDCG@10 = 0.749 on SciFact).

GeoPulse: Geopolitical Risk Intelligence Platform [GitHub](#)

PyTorch, Transformers, Graph Neural Networks, TimescaleDB, pgvector

- Built an end-to-end geopolitical risk intelligence platform on GDELT event data, combining a multi-task Transformer risk scorer, temporal forecasting models, and graph-based spillover analysis to monitor emerging regional instability.
- Developed a FastAPI- and Streamlit-based analytics system with explainability, geospatial visualization, and retrieval-augmented advisory generation, evaluated using 53-fold walk-forward backtesting across 210 countries.

Flowify: Code Comprehension via Semantic Graphs [GitHub](#)

Python, GraphRAG, NetworkX, FastAPI, MCP

- Built a GraphRAG-powered code exploration system that transforms repositories into semantic knowledge graphs and answers developer questions using graph-grounded retrieval rather than vector search alone.
- Implemented multi-language AST analysis, change-impact assessment, and MCP-compatible APIs, enabling LLM agents to navigate and reason over codebase structure.

Research & Patents

Master's Thesis: Multimodal AI for Clinical Voice Diagnostics [GitHub](#)

FAU Erlangen-Nürnberg | Indiana University Bloomington

Built a multimodal sequence-labeling model (Temporal Convolutional Network and Transformer encoder with cross-modal attention) fusing synchronized audio and glottal-area waveform signals to detect sparse vocal-fold events; F1 0.76, 0.60 IoU event localization, 0.86 patient-level diagnostic accuracy; integrated into a Python-based clinical application.

US Patent Applications (*HighRadius*): [US20250217746A1](#), ML-based prediction of financial transaction patterns · [US20240354786A1](#), ML systems for cash-flow forecasting.

Awards

3rd Prize, Healthcare Hackathon (Siemens Healthineers × Medical Valley): built *Flora*, a patient-facing conversational avatar with RAG over an embedded vector database of medical information, supporting medication reminders, appointment tracking, and PDF report summarization.

2nd Prize, Graphathon (Siemens Healthineers × Neo4j): built a job-market intelligence pipeline using LLM-based extraction of skills and technologies from live postings into Neo4j and Databricks, surfaced via a Streamlit dashboard.

Technical Skills

- **Domains:** Generative AI, Agentic Systems, NLP, Retrieval & RAG, Computer Vision, Multimodal Learning, Time-Series Forecasting, MLOps, Information Retrieval
- **Languages & Tools:** Python, SQL, TypeScript, Java, JavaScript, Bash, C/C++, Git
- **LLM & Agents:** LangChain, LangGraph, MCP, RAG pipelines, context engineering, Pydantic, OpenAI SDK, Anthropic SDK, Hugging Face Transformers, Ollama, fine-tuning (LoRA/PEFT)
- **Evaluation & Observability:** LangSmith, Pytest, Weights & Biases, MLFlow, Ragas
- **Data & Storage:** PostgreSQL/pgvector, ChromaDB, FAISS, Pinecone, Neo4j, MongoDB, Snowflake, Databricks, PySpark
- **Serving & MLOps:** FastAPI, Docker, Kubernetes, Streamlit, ONNX, CI/CD (Azure DevOps, Jenkins), Slurm
- **ML/DL Frameworks:** PyTorch, TensorFlow, scikit-learn
- **Cloud:** AWS (SageMaker, S3), Azure (Azure ML), GCP (BigQuery, Vertex AI, ADK)

Languages

English (C2) · German (A2, actively improving) · Hindi (C2) · Bengali (C1)